

Answers

Q	Ans	Q	Ans	Q	Ans
1	a	2	d	3	b
4	c	5	a	6	d
7	b	8	c	9	d
10	a	11	c	12	b
13	d	14	a	15	c
16	a	17	c	18	d
19	b	20	a	21	c
22	d	23	b	24	a
25	c	26	a	27	b
28	c	29	d	30	a
31	b	32	d	33	a
34	c	35	b	36	a
37	c	38	b	39	d
40	a	41	c	42	b
43	d	44	a	45	c
46	d	47	a	48	c
49	b	50	d	51	a
52	c	53	b	54	d
55	b	56	a	57	c
58	d	59	a	60	d

Selected working

(30) The photoelectric effect shows that light energy arrives in discrete packets, or photons: $E = hf$. This is why (a) — individual photons — is correct.

(31) The scientific method is “an ordered series of steps to acquire knowledge based on experimental evidence”: observe, ask a question, form a hypothesis, test it, analyse, conclude.

(33)–(35) The textbook lists four sources of experiential knowledge: sensory perception (seeing, touching, hearing), introspection (examining your own thoughts and feelings), memory, and testimony. Written testimony is “usually more reliable” than oral sources.

(42) A black hole is a region where gravity is so strong that nothing, not even light, can escape it. The first direct image (of the galaxy M87) was released in 2019.

(43) Gravitational waves are ripples in the fabric of spacetime predicted by Einstein’s general relativity and first detected directly by LIGO in 2015. They are not sound, seismic or ocean waves.

(44) The Higgs boson, discovered at CERN in 2012, is the particle associated with the yet-unanswered origin-of-mass mechanism (the Higgs field).

(45) The James Webb Space Telescope was designed to observe the universe mainly in the infrared, which lets it see the most distant and earliest objects.

(53) $0.0045 = 4.5 \times 10^{-3}$: the decimal point moves three places to the right, and the mantissa must lie between 1 and 10.

(54) 602 000 000 000 000 000 000 000 (Avogadro’s number) has 23 zeros after 602, so 6.02×10^{23} .

(55) In 0.00450 the leading zeros only fix the position of the decimal point and are not significant. The significant figures are 4, 5 and 0 (the final zero is significant), so the count is 3.

(56)–(57) Accuracy means closeness to the true value; precision means the closeness of repeated readings to one another. Precise readings can still be inaccurate (e.g. a wrongly zeroed scale).

(47) Hundreds of careful readings: metrology is the science of measurement, while meteorology is the study of weather. The two words are a common trap.

(59) SI prefixes: mega = 10^6 , so 1 MW = 10^6 W. Similarly kilo = 10^3 , micro = 10^{-6} , giga = 10^9 .

(60) Reading a meniscus at eye level avoids parallax error, which is caused by viewing the scale from an angle rather than directly in line with it.